Ghazi Muhammad Abdullah

Domain: Artificial Intelligence & Machine Learning

Submitted to : Mr. Mohazin Zahid

REPORT OF DAY 8 (Project)

Introduction

This report summarizes the activities completed during the M-Tech Production & Marketing AI/ML Internship Program on **5th July 2026**. The primary focus of today's work was the successful development of the **Employee Attrition Predictor**, a Machine Learning-based desktop application developed using Python and Tkinter. The project integrates data preprocessing, model training, prediction functionality, and a professional graphical user interface to predict whether an employee is likely to leave a company.

Objectives

- Complete the development of the **Employee Attrition Predictor** desktop application.
- Implement data preprocessing and Machine Learning model training.
- Develop a professional graphical user interface using Tkinter.
- Integrate prediction functionality with the trained Machine Learning model.
- Test the application and ensure its functionality.

Task Description

The assigned task involved developing the **Employee Attrition Predictor**, a complete Python desktop application that predicts employee attrition using Machine Learning. The project required implementing data preprocessing techniques, training and evaluating multiple Machine Learning algorithms, selecting the best-performing model, and integrating it into a Tkinter-based graphical user interface. The application also included features such as input validation, prediction history, model saving and loading, model performance visualization, and complete project documentation.

Work Done

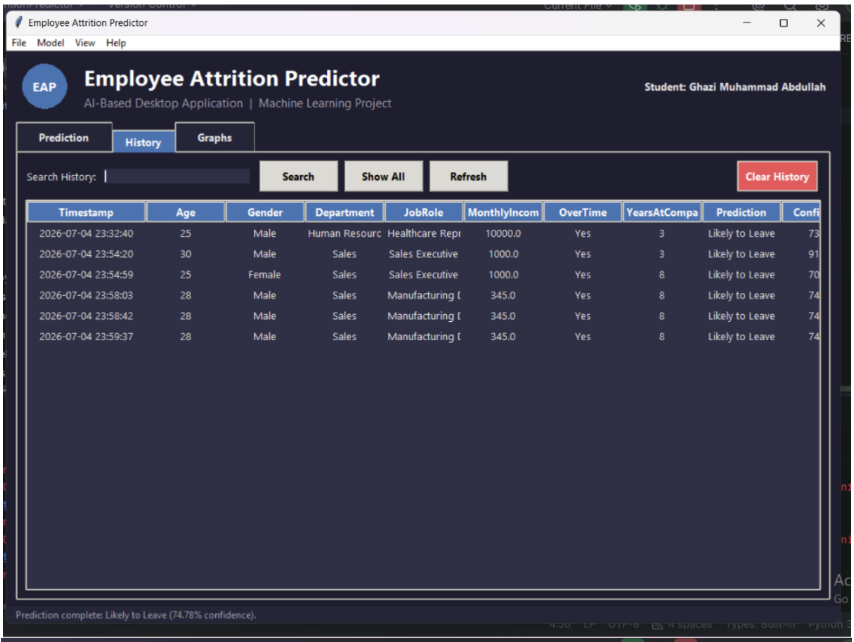
During today's work, I successfully developed the **Employee Attrition Predictor** desktop application from start to finish. I organized the project into multiple Python modules, making the code easier to manage and maintain. I prepared the employee dataset by cleaning the data, handling missing values, removing duplicate records, encoding categorical features, and scaling numerical values to prepare the data for Machine Learning.

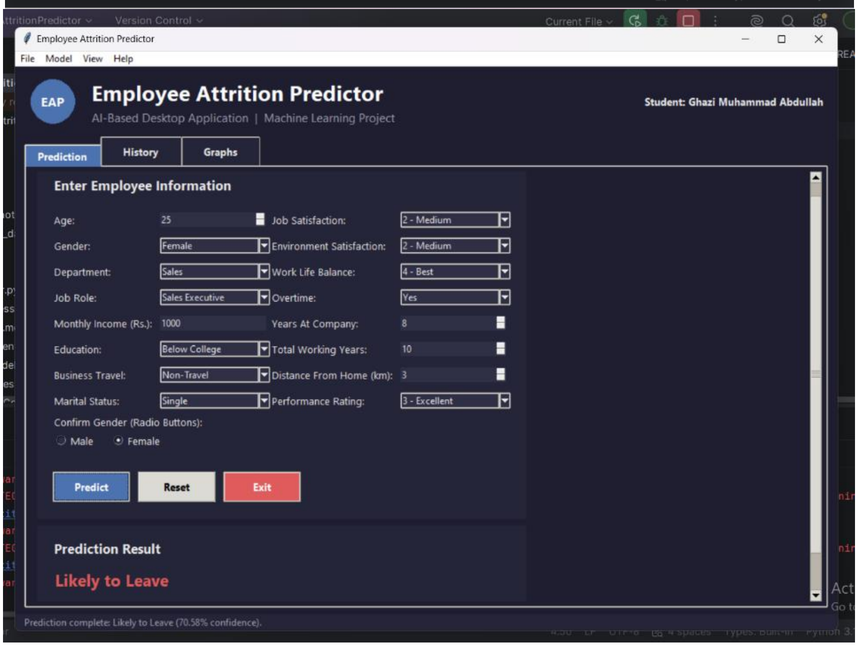
I trained and evaluated multiple Machine Learning algorithms, including **Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector**

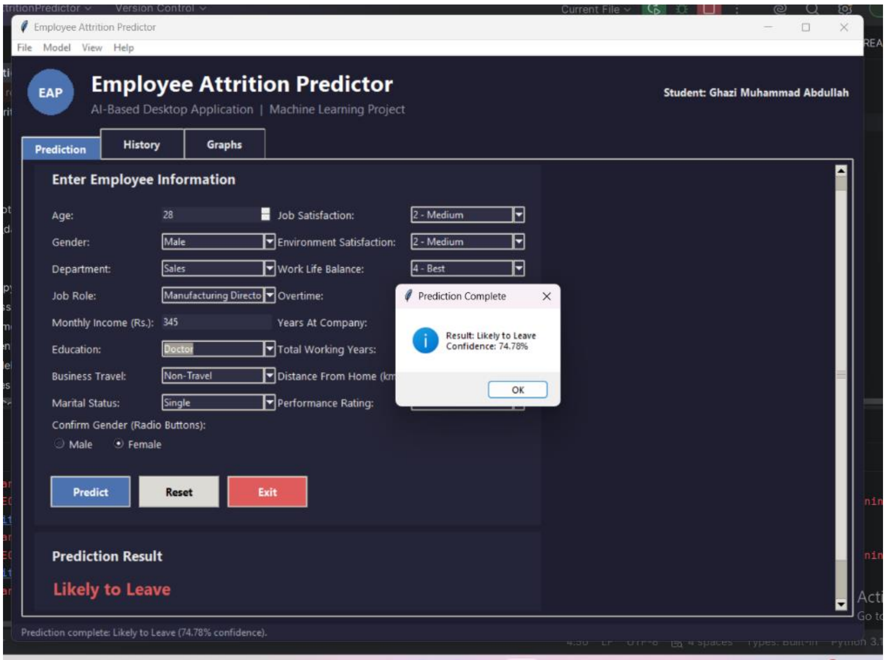
Machine. After comparing their performance, I selected the best-performing model and saved it using Joblib for future predictions.

I also designed a professional desktop interface using **Tkinter**, which included input fields, drop-down menus, prediction buttons, a menu bar, status bar, and proper input validation. The application was able to accept employee information, load the trained model, generate predictions, display the prediction result with confidence, and provide clear success or error messages.

In addition, I implemented features such as **Reset**, **Exit**, **Save Prediction History**, **View Prediction History**, **Search Previous Predictions**, and **Model Performance Graphs** to improve the usability of the application. Finally, I tested all modules individually and as a complete system to ensure the application worked correctly and efficiently.







Challenges Faced

Developing a complete Machine Learning desktop application required integrating several independent components into a single project. Managing data preprocessing, Machine Learning model training, and graphical user interface development while ensuring smooth communication between them required careful implementation and testing. Selecting the best-performing algorithm and correctly converting user input into a format suitable for prediction also required attention to detail. Through debugging and repeated testing, these challenges were successfully resolved.

Key Learnings

This project significantly improved my understanding of the complete Machine Learning development process. I gained practical experience in data preprocessing, feature engineering, model evaluation, and integrating trained models into a desktop application. I also strengthened my Python programming skills, learned how to organize a large project into multiple modules, and gained hands-on experience with Tkinter for developing professional desktop software.

Conclusion

Completing the **Employee Attrition Predictor** project provided valuable practical experience in Python programming, Machine Learning, and desktop application development. The project enhanced my understanding of the complete Machine Learning workflow, from data preprocessing and model training to prediction and graphical user interface development. The knowledge and experience gained through this project have strengthened my confidence in building AI-powered desktop applications and will support more advanced Artificial Intelligence and Machine Learning projects throughout the internship.

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Date: July 5th, 2026

Domain: AI/ML Intern

Intern Signature: